

MINI PROJECT REPORT

Multi-Subnet DHCP with Relay and Failover: Lease Dynamics and Exhaustion Analysis

Slot 5 — Multi-subnet DHCP with relay and failover: lease-dynamics and exhaustion analysis

Problem: Large networks run DHCP across subnets via relay and need resilience. Build relayed DHCP with failover and study lease behaviour. Objectives: set up Kea/ISC-DHCP pools across ≥ 3 subnets with relay agents; configure a failover peer; simulate pool exhaustion; measure lease-acquisition and failover-takeover time; capture DORA in Wireshark. Stack: Kea/ISC-DHCP, Containerlab, Wireshark · Type: Implementation · Deliverables: configs, DORA captures, failover-timing report

Submitted by

Akshya.B – 2117230040009

Anand.G - 2117230040010

B.E. Electronics and Communication Engineering

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ABSTRACT

This project presents an implementation-oriented study of Dynamic Host Configuration Protocol (DHCP) in a multi-subnet network. The environment was built on Ubuntu using Docker/Containerlab and the ISC DHCP Server. DHCP address pools were defined for multiple IP subnets and a relay-based architecture was used to support DHCP communication across subnet boundaries. Two DHCP servers were configured as a failover pair so that lease information and service availability could be studied. A deliberately small address pool was also prepared for lease-exhaustion analysis. DHCP client behaviour was observed from terminal output and packet-level traffic was captured with tcpdump and examined in Wireshark. The supplied evidence shows the Containerlab/Docker environment, ISC DHCP 4.4.1 operation, multi-subnet pool declarations, failover-peer configuration, DHCPDISCOVER/DHCPREQUEST activity, successful DHCPACK allocation, and a Wireshark view of DHCP broadcast traffic. The report documents the implemented architecture, configuration approach, testing procedure, observations, and limitations of the supplied experimental evidence.

1. INTRODUCTION

DHCP is a core network service used to automatically provide IP configuration to hosts. Instead of manually configuring every client, a DHCP server can allocate an address from a defined pool and provide parameters such as the subnet mask and default gateway. This becomes especially useful when the number of devices is large or when hosts frequently join and leave the network.

In a multi-subnet environment, a client broadcast normally does not cross a router. A DHCP relay agent solves this problem by receiving the client's DHCP message and forwarding it toward the centralized DHCP server. The server uses the relay information to determine which subnet should supply the address.

Availability is another important requirement. If only one DHCP server is available and it fails, new clients may be unable to obtain addresses. A failover pair provides a second DHCP server and a mechanism for coordinating lease information. This project combines these concepts in a containerized Ubuntu networking laboratory.

2. PROBLEM STATEMENT

Large IP networks commonly divide hosts into multiple subnets for scalability, security, and traffic management. DHCP clients normally begin communication using local broadcast messages, while a DHCP server may be located on another network. A relay agent is therefore required to forward DHCP messages between the client subnet and the centralized DHCP service. In addition, dependence on a single DHCP server creates an availability risk. A failover arrangement with a second server can improve service resilience and allow lease state to be coordinated between peers. This project addresses these issues by constructing a multi-subnet DHCP environment, introducing a relay, configuring a DHCP failover peer, and preparing a small pool to study lease allocation and exhaustion behaviour.

3. OBJECTIVES

- Configure DHCP address pools for multiple subnets.
- Implement DHCP relay functionality so clients on different network segments can reach the DHCP service.
- Configure a primary and secondary ISC DHCP server as a failover pair.
- Create a limited DHCP pool for observing lease behaviour and pool-exhaustion conditions.

- Observe the DHCP DORA sequence: DHCPDISCOVER, DHCPOFFER, DHCPREQUEST and DHCPACK.
- Capture DHCP traffic and inspect packet fields using tcpdump and Wireshark.
- Validate DHCP configuration and server operation using ISC DHCP diagnostic commands.

4. EXISTING AND PROPOSED SYSTEM

Existing/simple approach: In a basic network, DHCP may be provided by one server serving a single broadcast domain. This is easy to configure but does not demonstrate centralized service across multiple subnets and introduces a single-server dependency.

Proposed implementation: The project uses multiple DHCP scopes, a relay path, and two DHCP servers configured with a failover peer. A limited pool is used to examine lease availability and exhaustion. Packet capture provides visibility into the DHCP protocol at the network level.

5. TECHNOLOGIES AND TOOLS USED

Component	Specification
Operating System	Ubuntu 22.04
Container Platform	Docker
Network Lab	Containerlab
DHCP Server	ISC DHCP Server 4.4.1
DHCP Function	DHCP server with relay and failover configuration
Packet Capture	tcpdump and Wireshark
Transport Used by DHCP	UDP 67/68 for server/client communication
Failover Communication	TCP port 647 as configured in the project

6. SYSTEM ARCHITECTURE AND NETWORK CONFIGURATION

The implementation evidence shows a Linux/Ubuntu environment running Docker/Containerlab containers. The visible container list includes a DHCP relay, two client containers and a DHCP server container. The project configuration additionally defines a primary/secondary failover relationship.

The logical architecture is: DHCP clients on different subnets → DHCP relay → centralized DHCP service, with a failover peer between the two DHCP servers.

Subnet	DHCP Pool	Router
192.168.10.0/24	192.168.10.100 – 192.168.10.200	192.168.10.1
192.168.20.0/24	192.168.20.100 – 192.168.20.200	192.168.20.1

192.168.30.0/24

192.168.30.100 –
192.168.30.200

192.168.30.1

7. DHCP FUNDAMENTALS AND DORA PROCESS

DHCP uses a four-stage exchange commonly called DORA. The client sends DHCPDISCOVER, a server responds with DHCPOFFER, the client sends DHCPREQUEST, and the server confirms the lease with DHCPACK.

The supplied client evidence shows this exchange in operation. One observed allocation was 192.168.10.100. Another later exchange resulted in 192.168.10.109.

8. DHCP RELAY OPERATION

DHCP relay is required when the client and DHCP server are located in different IP subnets. The relay receives the client's broadcast request and forwards it toward the DHCP server, allowing centralized DHCP service to operate across routed network segments.

9. DHCP FAILOVER MECHANISM

The supplied failover configuration uses primary address 10.0.0.1, peer address 10.0.0.2 and TCP port 647. Parameters such as maximum response delay, maximum unacknowledged updates, MCLT, split and load-balance settings are visible in the configuration evidence.

The terminal evidence shows failover-related states and also records a configuration issue that was subsequently addressed before the DHCP daemon was started again.

10. LEASE DYNAMICS AND POOL EXHAUSTION

A DHCP lease is a temporary assignment of an IP address. As clients obtain leases, free addresses are consumed. Expired or released leases can return to the available pool.

For controlled exhaustion analysis, the project prepared the small range 192.168.20.100–192.168.20.102. The supplied evidence confirms this configuration, but it does not contain a complete quantitative exhaustion-time measurement.

11. IMPLEMENTATION METHODOLOGY

1. Create the Ubuntu project environment and Containerlab/Docker topology.
2. Start the DHCP relay, client and DHCP server containers.
3. Configure the DHCP server scopes for 192.168.10.0/24, 192.168.20.0/24 and 192.168.30.0/24.
4. Configure the primary and secondary failover relationship.
5. Configure the small pool for exhaustion analysis.
6. Validate the ISC DHCP configuration and start the server.
7. Run DHCP clients and observe DORA traffic.
8. Capture DHCP traffic with tcpdump and inspect the resulting packets in Wireshark.

12. CONFIGURATION AND VALIDATION

- ISC DHCP Server 4.4.1 was used.
- The configuration file `/etc/dhcp/dhcpd.conf` and lease database `/var/lib/dhcp/dhcpd.leases` were used in the server commands shown in the evidence.
- The server output shows listening and sending interfaces associated with the service/failover network.
- The three DHCP subnet ranges and the limited exhaustion pool were validated from the project evidence.

13. TESTING AND PACKET ANALYSIS

The client-side evidence shows DHCPDISCOVER messages sent from `eth1` to the broadcast address. Another test shows DHCPOFFER, DHCPREQUEST and DHCPACK messages, including successful allocation of `192.168.10.100` and `192.168.10.109`.

The Wireshark evidence shows DHCPDISCOVER frames with source IP `0.0.0.0`, destination IP `255.255.255.255`, UDP source port `68` and destination port `67`.

14. RESULTS AND DISCUSSION

Container deployment: The supplied terminal evidence shows four active containers: a DHCP relay, two clients, and a DHCP server.

DHCP server version: The server output identifies ISC DHCP Server 4.4.1.

Multi-subnet configuration: The evidence includes pools for `192.168.10.0/24`, `192.168.20.0/24` and `192.168.30.0/24`.

Failover addressing: The failover configuration uses `10.0.0.1` as the primary address, `10.0.0.2` as the peer address, and port `647`.

Lease-exhaustion pool: A deliberately small range, `192.168.20.100–192.168.20.102`, was prepared for exhaustion analysis.

DHCP exchange: Client output shows DHCPDISCOVER, DHCPOFFER, DHCPREQUEST and DHCPACK activity, with successful address allocation.

Packet capture: Wireshark shows DHCPDISCOVER traffic from `0.0.0.0` to `255.255.255.255` using UDP `68` to `67`.

Validation: The ISC DHCP daemon was started/tested using `dhcpd` configuration commands and lease-file options.

15. ADVANTAGES, LIMITATIONS AND APPLICATIONS

Advantages

- Centralized IP address management across multiple subnets.
- Relay-based support for clients outside the DHCP server's local broadcast domain.
- Redundant DHCP service through a failover-peer configuration.
- Controlled pool-exhaustion testing.

- Packet-level analysis through tcpdump and Wireshark.
- Containerized implementation without requiring physical networking hardware.

Limitations of the supplied evidence

- The supplied evidence does not contain a numerical lease-acquisition-time table or failover-takeover-time measurement.
- The screenshots show failover configuration and server-state messages, but they do not by themselves prove a completed server-failure/takeover experiment.
- The supplied Wireshark screenshot visibly contains DHCPDISCOVER packets; a complete DORA sequence is better demonstrated with a capture containing OFFER, REQUEST and ACK packets as well.
- The report therefore describes the observed results without inventing timing values or packet counts that are not visible in the supplied evidence.

Applications

- Enterprise and campus network address management.
- Virtualized networking laboratories.
- DHCP resilience testing.
- Lease-exhaustion analysis.
- Network troubleshooting and packet analysis.

16. FUTURE SCOPE

The implementation can be extended by collecting precise lease-acquisition time, failover-takeover time and recovery time using synchronized timestamps. A larger set of clients can be introduced to generate controlled load and produce lease-utilization graphs.

The lab can also be extended to include additional routed subnets, automated failure injection, periodic lease-database analysis, longer Wireshark captures, and automated scripts that summarize DHCP state transitions. For a stronger 5G-oriented demonstration, the DHCP service can be placed within a broader virtualized IP transport/service-management laboratory while keeping the DHCP experiment clearly separated from the 5G control-plane protocols.

17. CONCLUSION

The project demonstrates the practical configuration of DHCP across multiple subnets using a relay and a failover-oriented server arrangement. The supplied evidence shows a working Ubuntu containerized environment, ISC DHCP 4.4.1, three DHCP subnet scopes, a small exhaustion-analysis pool, failover-peer parameters, DHCP client exchanges and Wireshark packet inspection.

The experiment provides a useful understanding of how DHCP moves from simple single-subnet address allocation to a more resilient multi-subnet service. The combination of relay forwarding, centralized pools, lease management and a redundant server architecture is applicable to larger IP networks where automated address allocation and service availability are important.

18. REFERENCES

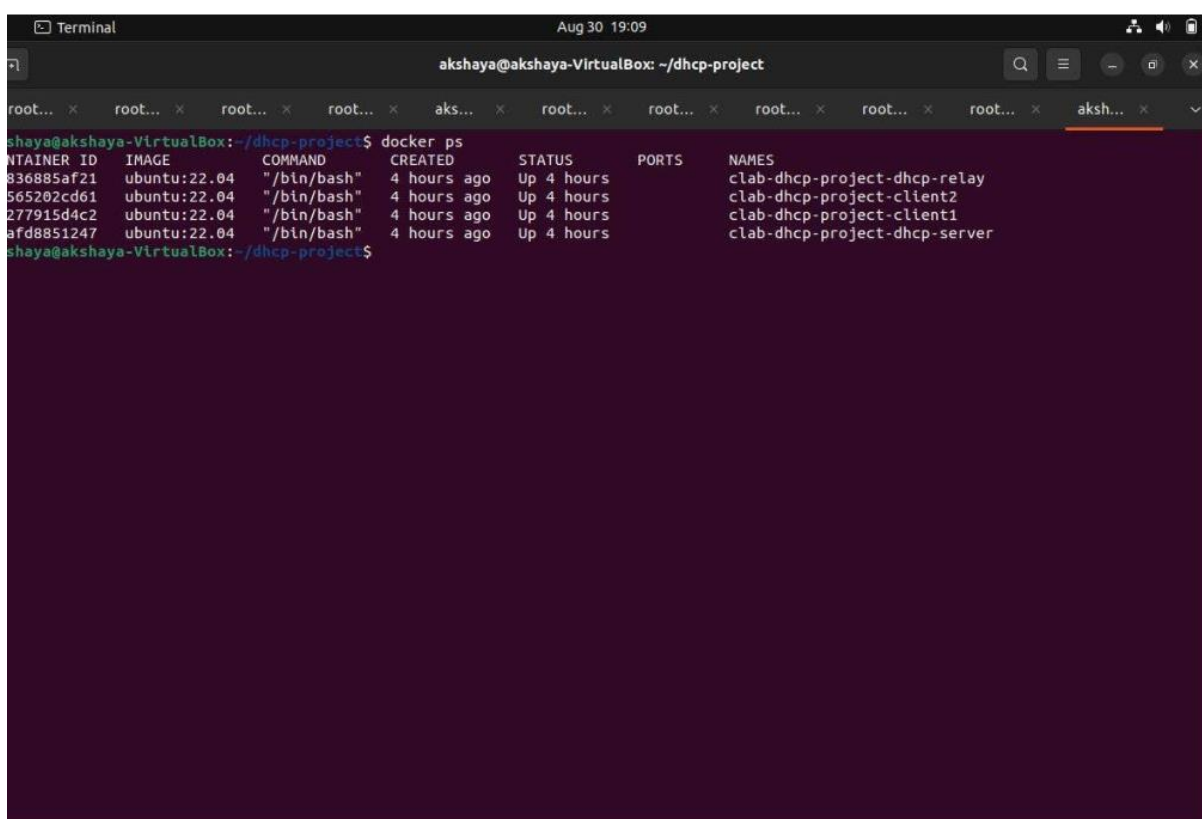
1. R. Droms, "Dynamic Host Configuration Protocol," RFC 2131, Internet Engineering Task Force.

2. Internet Systems Consortium, ISC DHCP Server documentation and configuration references.
3. Containerlab documentation and networking laboratory concepts.
4. Docker documentation for container execution and networking.
5. Wireshark documentation and DHCP packet-analysis references.

APPENDIX A – IMPLEMENTATION EVIDENCE

The following 13 figures reproduce the supplied implementation evidence.

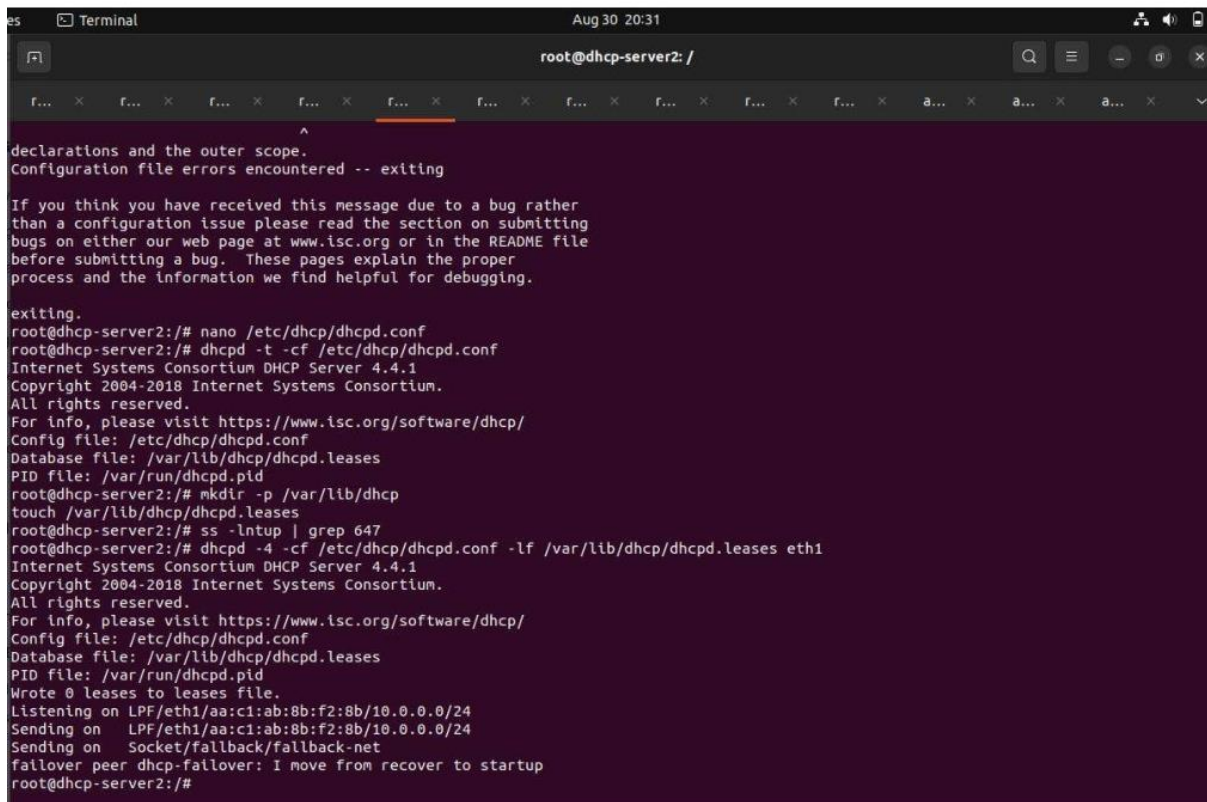
Figure A.1: Containerlab/Docker environment showing the DHCP relay, two clients, and DHCP server containers.



A terminal window titled "Terminal" with a timestamp of "Aug 30 19:09". The prompt is "akshaya@akshaya-VirtualBox: ~/dhcp-project". The user has executed the command "docker ps", which displays a table of running containers. The table has columns for CONTAINER ID, IMAGE, COMMAND, CREATED, STATUS, PORTS, and NAMES. There are four containers listed: a DHCP relay, two clients, and a DHCP server, all based on the ubuntu:22.04 image and running the /bin/bash command. They were all created 4 hours ago and are currently in an "Up" status.

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
836885af21	ubuntu:22.04	"/bin/bash"	4 hours ago	Up 4 hours		clab-dhcp-project-dhcp-relay
565202cd61	ubuntu:22.04	"/bin/bash"	4 hours ago	Up 4 hours		clab-dhcp-project-client2
277915d4c2	ubuntu:22.04	"/bin/bash"	4 hours ago	Up 4 hours		clab-dhcp-project-client1
afdb8851247	ubuntu:22.04	"/bin/bash"	4 hours ago	Up 4 hours		clab-dhcp-project-dhcp-server

Figure A.2: ISC DHCP Server 4.4.1 startup on the secondary/failover-side container and its failover state messages.



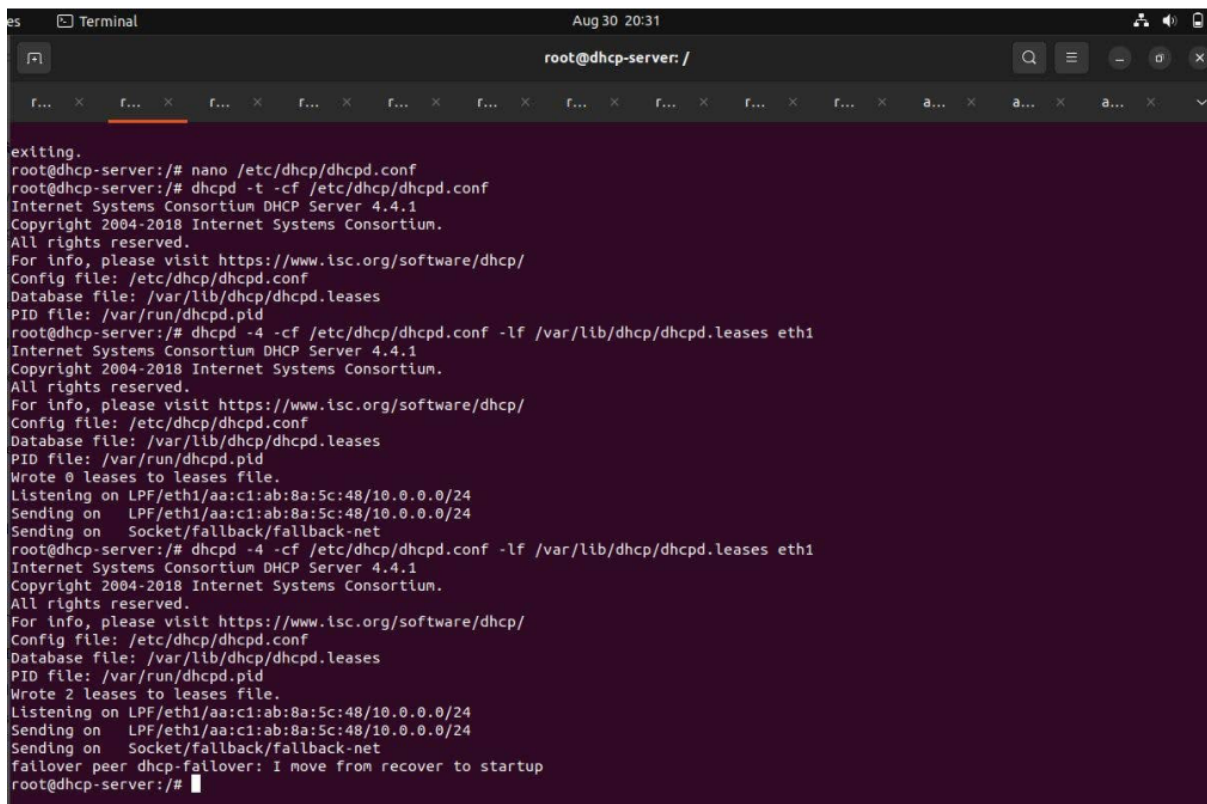
```
es  Terminal  Aug 30 20:31  root@dhcp-server2: /

declarations and the outer scope.
Configuration file errors encountered -- exiting

If you think you have received this message due to a bug rather
than a configuration issue please read the section on submitting
bugs on either our web page at www.isc.org or in the README file
before submitting a bug. These pages explain the proper
process and the information we find helpful for debugging.

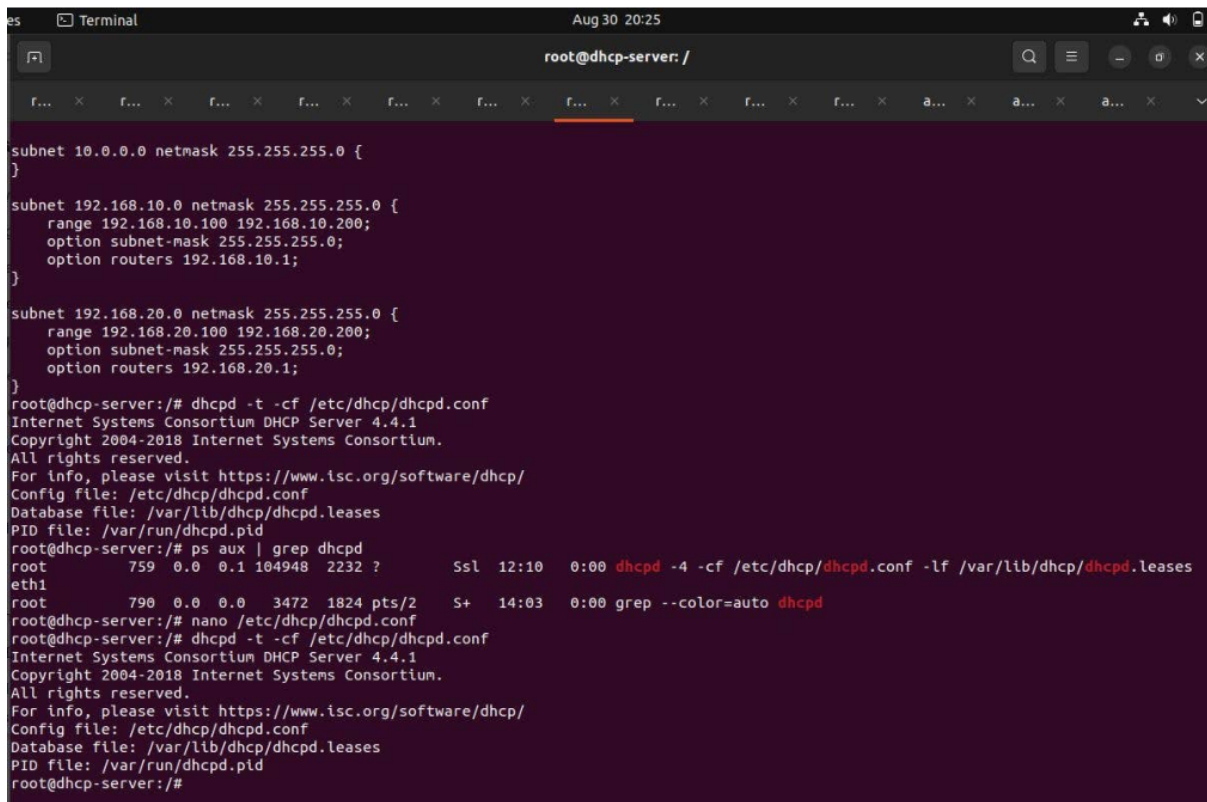
exiting.
root@dhcp-server2:/# nano /etc/dhcp/dhcpd.conf
root@dhcp-server2:/# dhcpd -t -cf /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
root@dhcp-server2:/# mkdir -p /var/lib/dhcp
touch /var/lib/dhcp/dhcpd.leases
root@dhcp-server2:/# ss -lntup | grep 647
root@dhcp-server2:/# dhcpd -4 -cf /etc/dhcp/dhcpd.conf -lf /var/lib/dhcp/dhcpd.leases eth1
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Wrote 0 leases to leases file.
Listening on LPF/eth1/aa:c1:ab:8b:f2:8b/10.0.0.0/24
Sending on   LPF/eth1/aa:c1:ab:8b:f2:8b/10.0.0.0/24
Sending on   Socket/fallback/fallback-net
failover peer dhcp-failover: I move from recover to startup
root@dhcp-server2:/#
```

Figure A.3: ISC DHCP Server 4.4.1 startup on the primary DHCP server.

A terminal window titled 'Terminal' with a timestamp of 'Aug 30 20:31'. The prompt is 'root@dhcp-server: /'. The user enters 'exitting.' followed by 'nano /etc/dhcp/dhcpd.conf'. Then they run 'dhcpd -t -cf /etc/dhcp/dhcpd.conf', which shows the server version (4.4.1), copyright (2004-2018), and config files. Next, they run 'dhcpd -4 -cf /etc/dhcp/dhcpd.conf -lf /var/lib/dhcp/dhcpd.leases eth1', which shows the server version, copyright, config files, and that it wrote 0 leases. Finally, they run 'dhcpd -4 -cf /etc/dhcp/dhcpd.conf -lf /var/lib/dhcp/dhcpd.leases eth1' again, which shows the server version, copyright, config files, and that it wrote 2 leases. The terminal output shows the server listening on LPF/eth1/aa:c1:ab:8a:5c:48/10.0.0.0/24 and sending on LPF/eth1/aa:c1:ab:8a:5c:48/10.0.0.0/24 and Socket/fallback/fallback-net. The last line shows 'fallover peer dhcp-fallover: 1 move from recover to startup' and the prompt 'root@dhcp-server: /#'.

```
exitting.
root@dhcp-server:/# nano /etc/dhcp/dhcpd.conf
root@dhcp-server:/# dhcpd -t -cf /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
root@dhcp-server:/# dhcpd -4 -cf /etc/dhcp/dhcpd.conf -lf /var/lib/dhcp/dhcpd.leases eth1
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Wrote 0 leases to leases file.
Listening on LPF/eth1/aa:c1:ab:8a:5c:48/10.0.0.0/24
Sending on LPF/eth1/aa:c1:ab:8a:5c:48/10.0.0.0/24
Sending on Socket/fallback/fallback-net
root@dhcp-server:/# dhcpd -4 -cf /etc/dhcp/dhcpd.conf -lf /var/lib/dhcp/dhcpd.leases eth1
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Wrote 2 leases to leases file.
Listening on LPF/eth1/aa:c1:ab:8a:5c:48/10.0.0.0/24
Sending on LPF/eth1/aa:c1:ab:8a:5c:48/10.0.0.0/24
Sending on Socket/fallback/fallback-net
fallover peer dhcp-fallover: 1 move from recover to startup
root@dhcp-server:/#
```

Figure A.4: Multi-subnet DHCP configuration showing address ranges and router options for the configured networks.



```
es  Terminal  Aug 30 20:25
root@dhcp-server: /

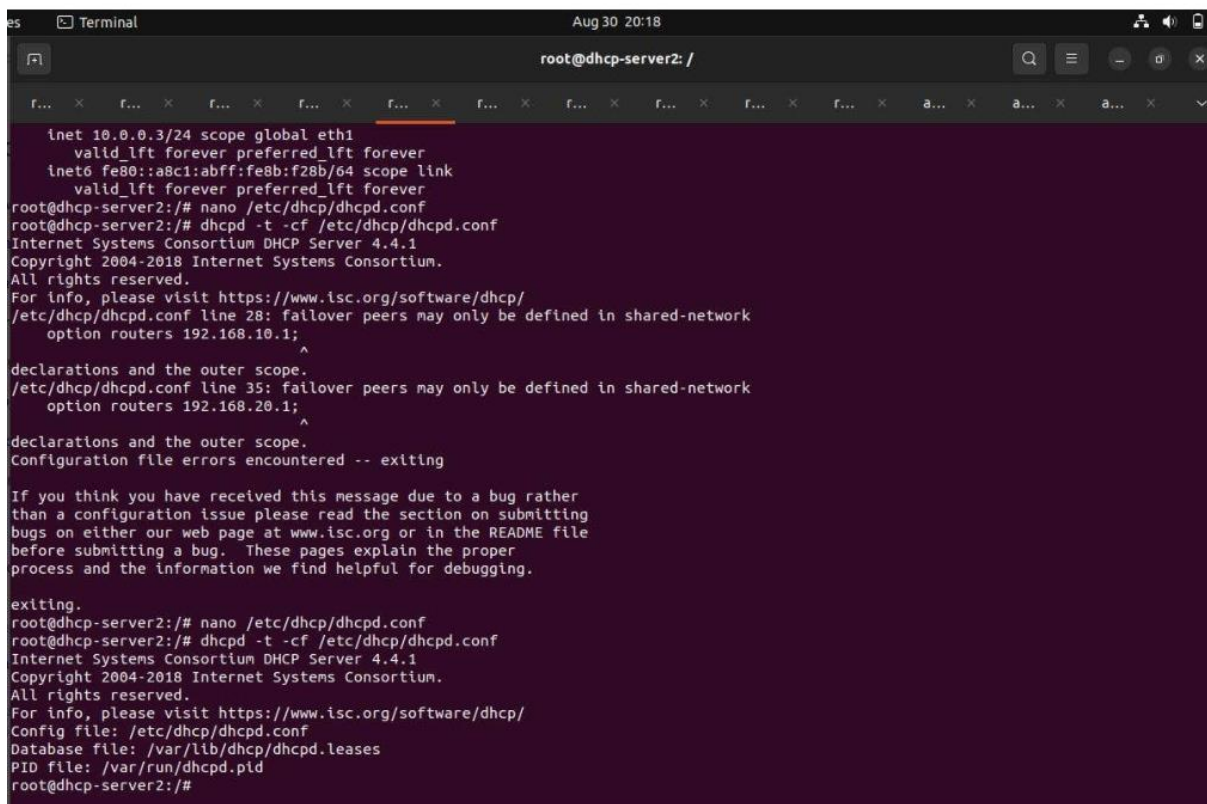
subnet 10.0.0.0 netmask 255.255.255.0 {
}

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.100 192.168.10.200;
    option subnet-mask 255.255.255.0;
    option routers 192.168.10.1;
}

subnet 192.168.20.0 netmask 255.255.255.0 {
    range 192.168.20.100 192.168.20.200;
    option subnet-mask 255.255.255.0;
    option routers 192.168.20.1;
}

root@dhcp-server: /# dhcpd -t -cf /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
root@dhcp-server: /# ps aux | grep dhcpd
root      759  0.0  0.1 104948 2232 ?        Ssl  12:10   0:00 dhcpd -4 -cf /etc/dhcp/dhcpd.conf -lf /var/lib/dhcp/dhcpd.leases
eth1
root      790  0.0  0.0   3472 1824 pts/2    S+   14:03   0:00 grep --color=auto dhcpd
root@dhcp-server: /# nano /etc/dhcp/dhcpd.conf
root@dhcp-server: /# dhcpd -t -cf /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
root@dhcp-server: /#
```

Figure A.5: DHCP configuration validation and failover-related configuration troubleshooting on the secondary server.



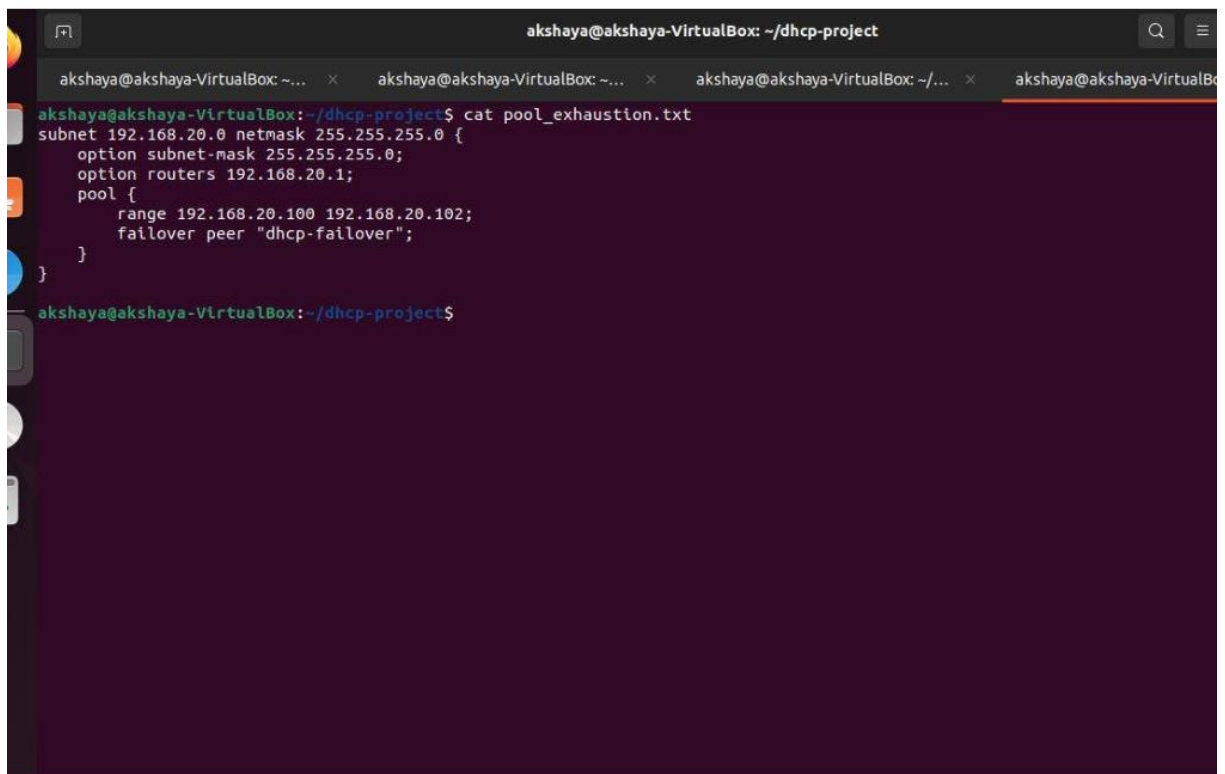
```
es Terminal Aug 30 20:18
root@dhcp-server2: /

inet 10.0.0.3/24 scope global eth1
    valid_lft forever preferred_lft forever
inet6 fe80::a8c1:abff:fe8b:f28b/64 scope link
    valid_lft forever preferred_lft forever
root@dhcp-server2:/# nano /etc/dhcp/dhcpd.conf
root@dhcp-server2:/# dhcpd -t -cf /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
/etc/dhcp/dhcpd.conf line 28: failover peers may only be defined in shared-network
    option routers 192.168.10.1;
    ^
declarations and the outer scope.
/etc/dhcp/dhcpd.conf line 35: failover peers may only be defined in shared-network
    option routers 192.168.20.1;
    ^
declarations and the outer scope.
Configuration file errors encountered -- exiting

If you think you have received this message due to a bug rather
than a configuration issue please read the section on submitting
bugs on either our web page at www.isc.org or in the README file
before submitting a bug. These pages explain the proper
process and the information we find helpful for debugging.

exiting.
root@dhcp-server2:/# nano /etc/dhcp/dhcpd.conf
root@dhcp-server2:/# dhcpd -t -cf /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
root@dhcp-server2:/#
```

Figure A.6: Small DHCP pool prepared for lease-exhaustion analysis: 192.168.20.100–192.168.20.102.

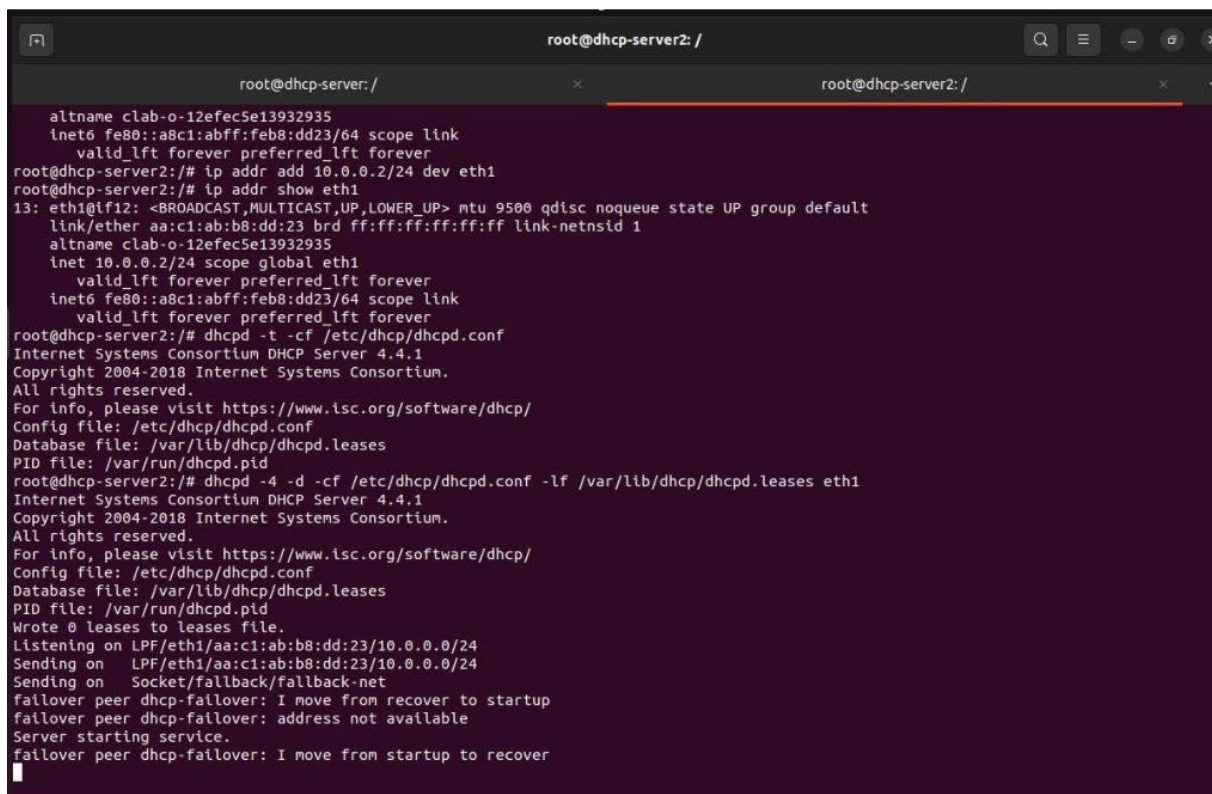
A terminal window titled 'akshaya@akshaya-VirtualBox: ~/dhcp-project' with four tabs. The active tab shows the command 'cat pool_exhaustion.txt' and its output, which is a DHCP configuration snippet. The configuration defines a subnet 192.168.20.0 with a netmask of 255.255.255.0, a router at 192.168.20.1, and a DHCP pool with a range of 192.168.20.100 to 192.168.20.102 and a fallback to 'dhcp-fallover'.

```
akshaya@akshaya-VirtualBox: ~/dhcp-project
akshaya@akshaya-VirtualBox: ~/dhcp-project$ cat pool_exhaustion.txt
subnet 192.168.20.0 netmask 255.255.255.0 {
    option subnet-mask 255.255.255.0;
    option routers 192.168.20.1;
    pool {
        range 192.168.20.100 192.168.20.102;
        fallover peer "dhcp-fallover";
    }
}
akshaya@akshaya-VirtualBox: ~/dhcp-project$
```

Figure A.7: Subnet evidence file showing the three configured DHCP subnet ranges.

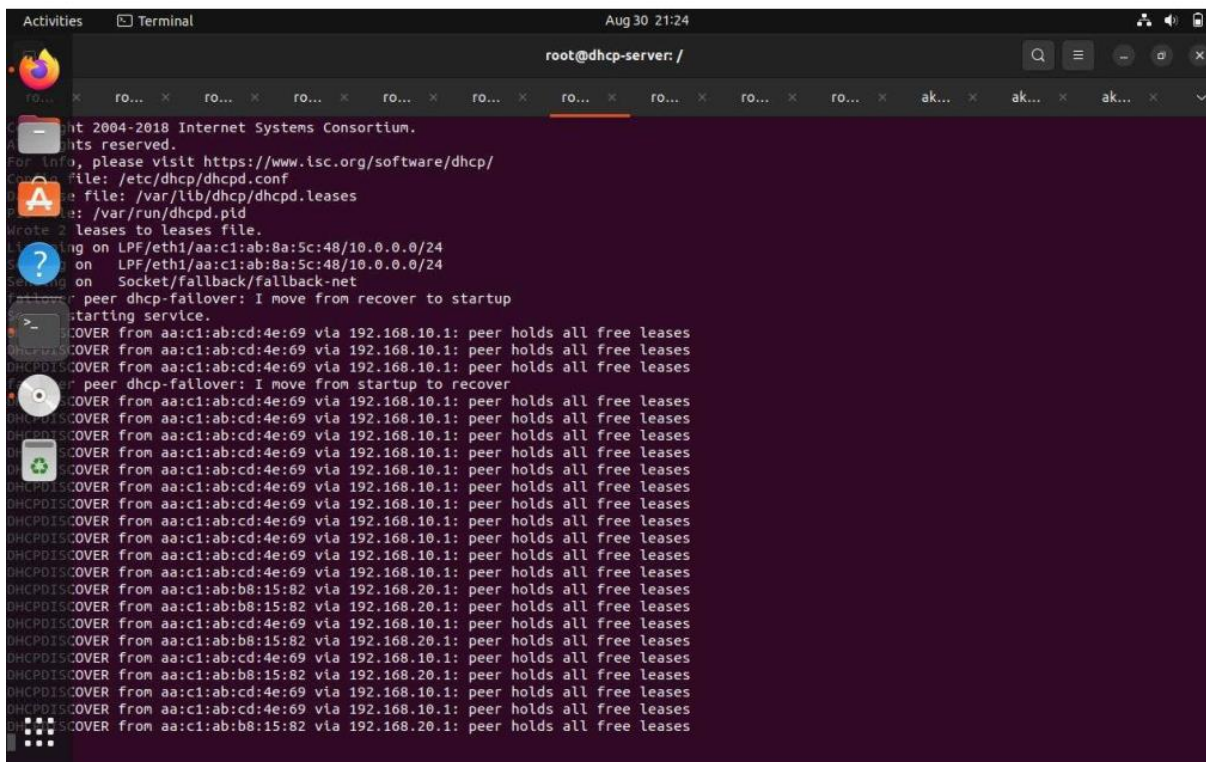
```
akshaya@akshaya-VirtualBox:~/dhcp-project$ cat subnet_evidence.txt
subnet 192.168.10.0 netmask 255.255.255.0 {
subnet 192.168.20.0 netmask 255.255.255.0 {
    range 192.168.20.100 192.168.20.102;
subnet 192.168.30.0 netmask 255.255.255.0 {
    range 192.168.30.100 192.168.30.200;
akshaya@akshaya-VirtualBox:~/dhcp-project$
```

Figure A.8: Secondary DHCP server interface addressing and failover-related service output.



```
root@dhcp-server2: /
altname clab-o-12efec5e13932935
inet6 fe80::a8c1:abff:feb8:dd23/64 scope link
valid_lft forever preferred_lft forever
root@dhcp-server2: /# ip addr add 10.0.0.2/24 dev eth1
root@dhcp-server2: /# ip addr show eth1
13: eth1@if12: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9500 qdisc noqueue state UP group default
    link/ether aa:c1:ab:b8:dd:23 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    altname clab-o-12efec5e13932935
    inet 10.0.0.2/24 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::a8c1:abff:feb8:dd23/64 scope link
        valid_lft forever preferred_lft forever
root@dhcp-server2: /# dhcpd -t -cf /etc/dhcp/dhcpd.conf
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
root@dhcp-server2: /# dhcpd -4 -d -cf /etc/dhcp/dhcpd.conf -lf /var/lib/dhcp/dhcpd.leases eth1
Internet Systems Consortium DHCP Server 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/
Config file: /etc/dhcp/dhcpd.conf
Database file: /var/lib/dhcp/dhcpd.leases
PID file: /var/run/dhcpd.pid
Wrote 0 leases to leases file.
Listening on LPF/eth1/aa:c1:ab:b8:dd:23/10.0.0.0/24
Sending on   LPF/eth1/aa:c1:ab:b8:dd:23/10.0.0.0/24
Sending on   Socket/fallback/fallback-net
failover peer dhcp-failover: I move from recover to startup
failover peer dhcp-failover: address not available
Server starting service.
failover peer dhcp-failover: I move from startup to recover
```

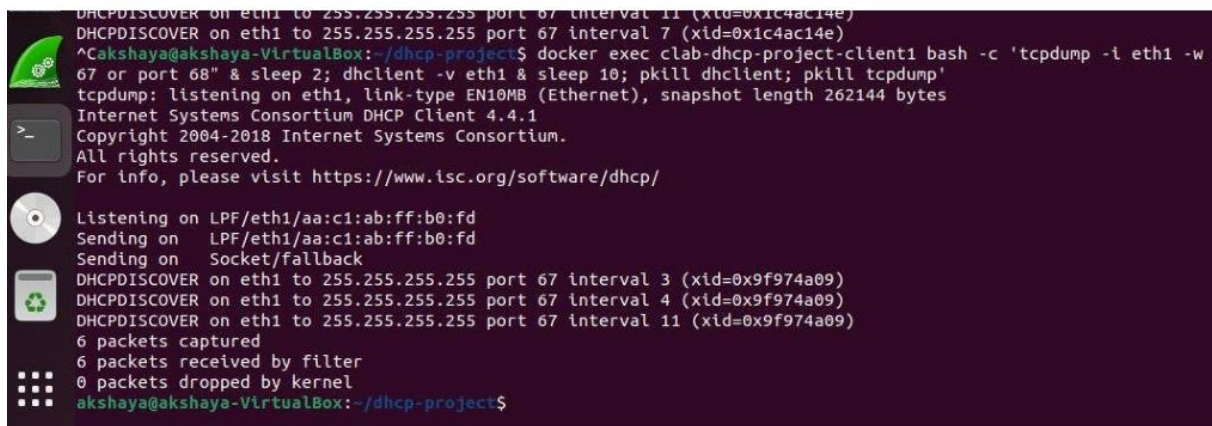
Figure A.9: DHCP server terminal output showing repeated DHCPDISCOVER traffic arriving through the configured network.



The screenshot shows a terminal window titled "Terminal" with the prompt "root@dhcp-server: /". The terminal output displays the following sequence of events:

```
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All rights reserved.  
For info, please visit https://www.isc.org/software/dhcp/  
config file: /etc/dhcp/dhcpd.conf  
lease file: /var/lib/dhcp/dhcpd.leases  
pid file: /var/run/dhcpd.pid  
wrote 2 leases to leases file.  
Listening on LPF/eth1/aa:c1:ab:cd:4e:69/192.168.10.1: peer holds all free leases  
Listening on LPF/eth1/aa:c1:ab:cd:4e:69/192.168.10.1: peer holds all free leases  
Selecting on Socket/fallback/fallback-net  
peer dhcp-fallover: I move from recover to startup  
starting service.  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
peer dhcp-fallover: I move from startup to recover  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:b8:15:82 via 192.168.20.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:b8:15:82 via 192.168.20.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:b8:15:82 via 192.168.20.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:b8:15:82 via 192.168.20.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:cd:4e:69 via 192.168.10.1: peer holds all free leases  
DHCPDISCOVER from aa:c1:ab:b8:15:82 via 192.168.20.1: peer holds all free leases
```

Figure A.10: Client-side tcpdump capture showing DHCPDISCOVER packets transmitted on eth1 toward the broadcast address.



```
DHCPDISCOVER on eth1 to 255.255.255.255 port 67 interval 11 (xid=0x1c4ac14e)
DHCPDISCOVER on eth1 to 255.255.255.255 port 67 interval 7 (xid=0x1c4ac14e)
^Cakshaya@akshaya-VirtualBox:~/dhcp-project$ docker exec clab-dhcp-project-client1 bash -c 'tcpdump -i eth1 -w
67 or port 68" & sleep 2; dhclient -v eth1 & sleep 10; pkill dhclient; pkill tcpdump'
tcpdump: listening on eth1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
Internet Systems Consortium DHCP Client 4.4.1
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All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/eth1/aa:c1:ab:ff:b0:fd
Sending on LPF/eth1/aa:c1:ab:ff:b0:fd
Sending on Socket/fallback
DHCPDISCOVER on eth1 to 255.255.255.255 port 67 interval 3 (xid=0x9f974a09)
DHCPDISCOVER on eth1 to 255.255.255.255 port 67 interval 4 (xid=0x9f974a09)
DHCPDISCOVER on eth1 to 255.255.255.255 port 67 interval 11 (xid=0x9f974a09)
6 packets captured
6 packets received by filter
0 packets dropped by kernel
akshaya@akshaya-VirtualBox:~/dhcp-project$
```

Figure A.11: Client-side DHCP exchange showing DHCPDISCOVER, DHCPOFFER, DHCPREQUEST and DHCPACK, including successful allocation of 192.168.10.100 and later 192.168.10.109.

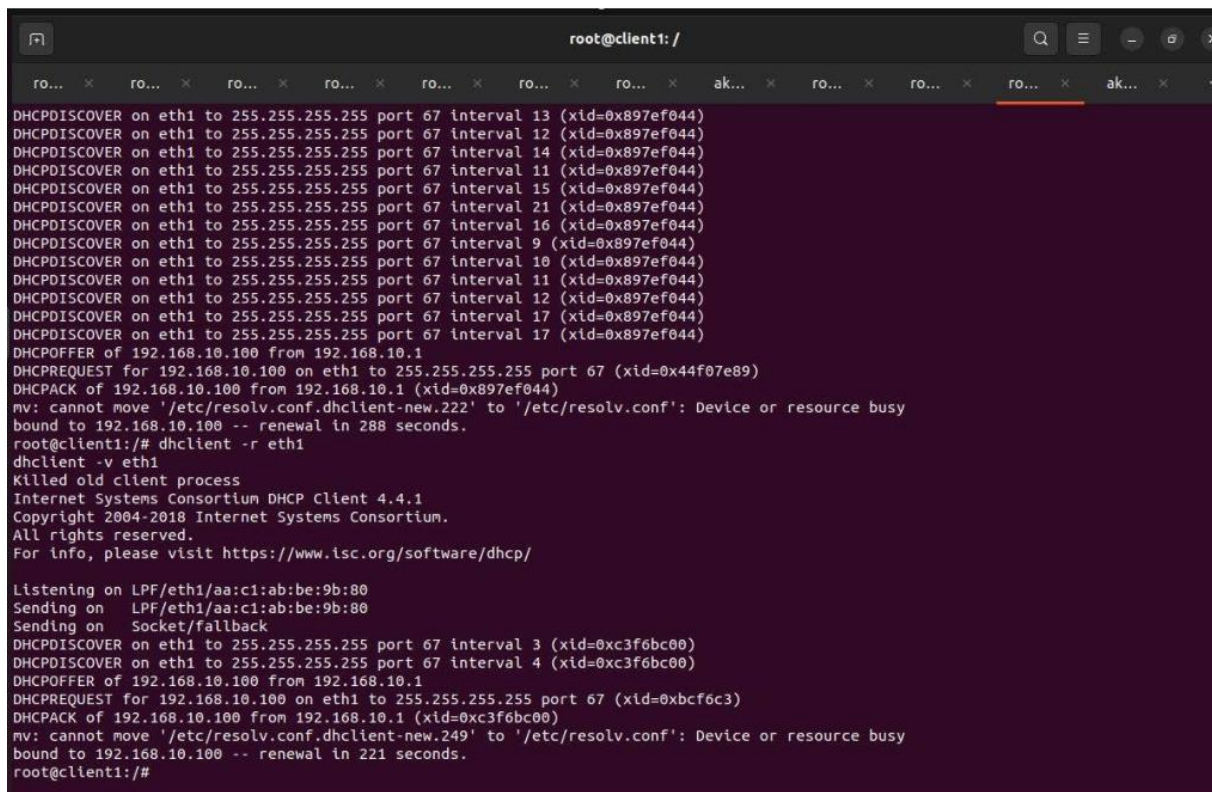
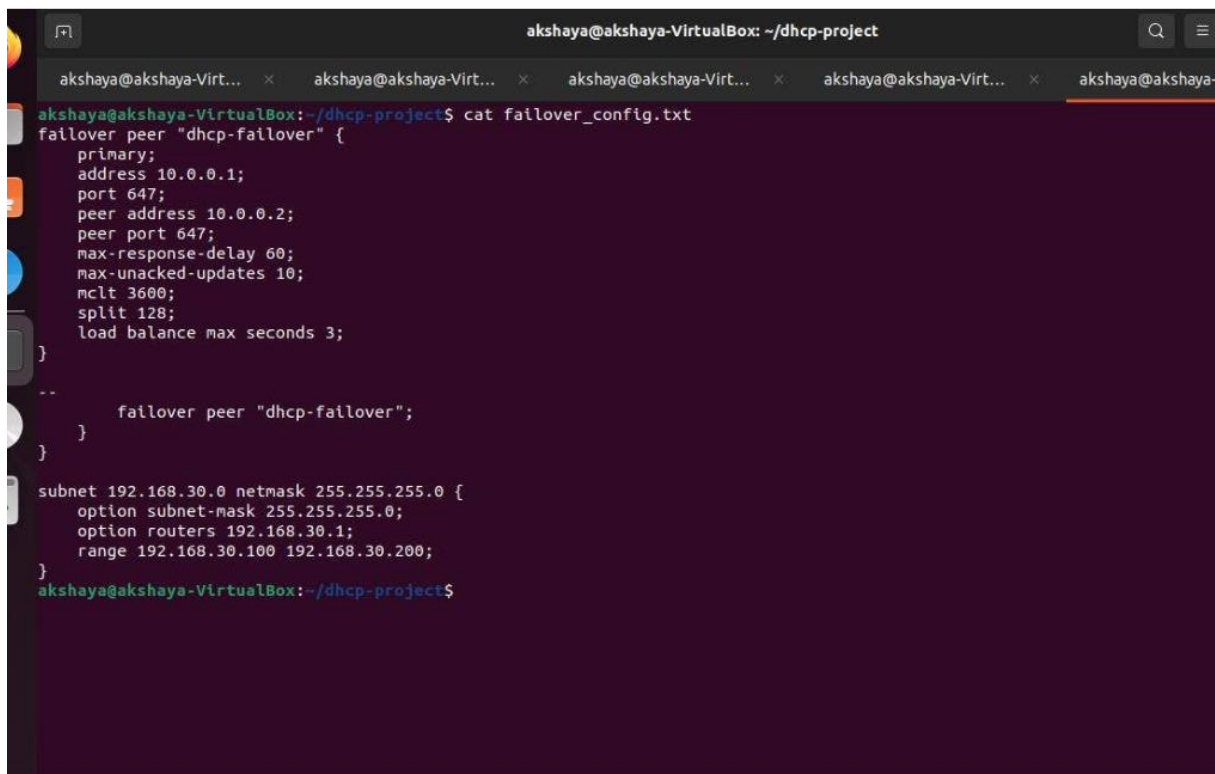
A terminal window titled 'root@client1: /' with a dark purple background and white text. The window shows a series of DHCP-related log messages. The first part shows multiple 'DHCPDISCOVER' messages on eth1 to 255.255.255.255 port 67 at various intervals. This is followed by a 'DHCPOFFER' for 192.168.10.100 from 192.168.10.1, a 'DHCPREQUEST' for the same IP, and a 'DHCPACK' for 192.168.10.100 from 192.168.10.1. Then, an error message appears: 'mv: cannot move '/etc/resolv.conf.dhclient-new.222' to '/etc/resolv.conf': Device or resource busy', followed by 'bound to 192.168.10.100 -- renewal in 288 seconds.'. The user then enters 'root@client1:/# dhclient -r eth1' and 'dhclient -v eth1'. The terminal shows 'Killed old client process', 'Internet Systems Consortium DHCP Client 4.4.1', 'Copyright 2004-2018 Internet Systems Consortium.', 'All rights reserved.', and 'For info, please visit https://www.isc.org/software/dhcp/'. This is followed by 'Listening on LPF/eth1/aa:c1:ab:be:9b:80', 'Sending on LPF/eth1/aa:c1:ab:be:9b:80', 'Sending on Socket/fallback', and another 'DHCPDISCOVER' on eth1 to 255.255.255.255 port 67 at interval 3. This is followed by another 'DHCPDISCOVER' at interval 4, a 'DHCPOFFER' for 192.168.10.100 from 192.168.10.1, a 'DHCPREQUEST' for 192.168.10.100 on eth1 to 255.255.255.255 port 67, and a 'DHCPACK' for 192.168.10.100 from 192.168.10.1. Another error message appears: 'mv: cannot move '/etc/resolv.conf.dhclient-new.249' to '/etc/resolv.conf': Device or resource busy', followed by 'bound to 192.168.10.100 -- renewal in 221 seconds.'. The prompt returns to 'root@client1:/#'.

Figure A.12: Failover-peer configuration showing primary address 10.0.0.1, peer address 10.0.0.2, and TCP port 647.

A terminal window titled 'akshaya@akshaya-VirtualBox: ~/dhcp-project' displays the contents of a file named 'failover_config.txt'. The configuration defines a failover peer named 'dhcp-failover' with a primary address of 10.0.0.1 and a peer address of 10.0.0.2, both using port 647. It also sets various timing parameters like max-response-delay, max-unacked-updates, mclt, split, and load balance. A subnet configuration for 192.168.30.0 is also shown at the bottom.

```
akshaya@akshaya-VirtualBox: ~/dhcp-project
akshaya@akshaya-VirtualBox:~/dhcp-project$ cat failover_config.txt
failover peer "dhcp-failover" {
    primary;
    address 10.0.0.1;
    port 647;
    peer address 10.0.0.2;
    peer port 647;
    max-response-delay 60;
    max-unacked-updates 10;
    mclt 3600;
    split 128;
    load balance max seconds 3;
}

--
    failover peer "dhcp-failover";
}

subnet 192.168.30.0 netmask 255.255.255.0 {
    option subnet-mask 255.255.255.0;
    option routers 192.168.30.1;
    range 192.168.30.100 192.168.30.200;
}
akshaya@akshaya-VirtualBox:~/dhcp-project$
```

Figure A.13: Wireshark capture of DHCPDISCOVER packets, including UDP source port 68 and destination port 67.

